

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	AGRI-ANALYTICS-001
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Crop Production Data Management	Upload crop production data (State/District/Year) Edit and update existing data Delete data records Data validation and error handling
FR-4	Data Analysis & Visualization	View crop production trends over years Compare crop production across states/districts Filter by crop type, state, year, season Visualizations: Charts, Graphs, Maps, Tables
FR-5	Reports & Export	Generate summary reports Export reports in PDF, Excel, CSV Print and share reports
FR-6	User Management	Admin login and dashboard Manage user roles and permissions View user activity logs

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system shall provide an intuitive and user-friendly interface for all users including farmers, analysts and administrators.
NFR-2	Security	The system shall ensure data security, user authentication, role-based access control and encryption of sensitive information.
NFR-3	Reliability	The system shall ensure accurate data processing and provide reliable results with minimal errors.
NFR-4	Performance	The system shall deliver fast response times for data retrieval, analysis and report generation.
NFR-5	Availability	The system shall be available 24/7 with minimum downtime for users across India.
NFR-6	Scalability	The system shall be designed to handle large volumes of data and support future enhancements and additional users.